

STANDARD AGCM, AB, Canada

WMO: 714400

Lat: 51.2283N

Lon: 112.9819W

Elev: 2956

StdP: 13.19

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-17.9	-13.2	-23.9	1.6	-17.2	-19.1	2.2	-13.0	28.2	31.0	25.1	35.0	6.5	190	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.8	84.9	61.3	81.6	60.4	78.4	59.6	65.4	78.2	63.6	76.1	61.8	74.0	9.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	89.4	70.9	58.8	82.6	69.4	56.8	76.9	67.2	32.0	78.1	30.4	76.5	29.1	73.8	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.6	21.8	19.2	DB	-24.3	91.6	4.3	3.6	-27.4	94.2	-29.9	96.4	-32.3	98.4	-35.5	101.0	(4)
(5)			WB	-24.6	70.9	4.3	2.7	-27.6	72.8	-30.1	74.4	-32.5	75.9	-35.6	77.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.3	19.3	16.7	27.5	39.9	52.5	58.6	63.7	62.6	54.9	41.4	28.0	16.6		
	DBStd	20.57	15.17	14.97	14.26	9.45	7.96	5.91	4.98	5.97	9.25	9.70	13.23	14.67		
	HDD50	5024	951	931	699	321	66	6	0	2	54	295	663	1036		
	HDD65	9143	1416	1351	1163	754	393	205	84	116	319	731	1110	1501		
	CDD50	1474	0	0	1	17	142	264	424	393	201	29	3	0		
	CDD65	119	0	0	0	0	4	13	44	42	16	0	0	0		
	CDH74	1861	0	0	0	12	139	195	569	605	331	10	0	0		
	CDH80	461	0	0	0	1	18	34	139	171	97	1	0	0		
(14) Wind	WSAvg	9.6	9.9	9.9	9.4	9.9	10.4	10.4	8.5	8.2	8.5	10.1	9.9	10.2		
(15) Precipitation	PrecAvg	15.1	0.6	0.4	0.6	1.0	1.9	3.0	2.6	2.1	1.3	0.6	0.6	0.5		
	PrecMax	19.3	1.3	0.9	2.1	2.4	3.7	6.0	5.9	4.3	3.2	1.4	1.9	1.1		
	PrecMin	9.0	0.0	0.0	0.1	0.1	0.2	1.3	0.9	0.1	0.2	0.2	0.1	0.0		
	PrecStd	2.6	0.3	0.2	0.4	0.6	0.9	1.2	1.2	1.1	0.7	0.4	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	48.6	61.3	73.0	82.5	84.4	88.5	89.4	87.5	73.8	63.1	49.8		
		MCWB	39.8	39.6	46.0	50.5	55.1	62.0	64.9	61.3	59.3	52.8	46.9	39.3		
	2%	DB	44.4	43.5	55.1	66.7	78.2	79.2	84.0	84.5	82.9	66.5	55.1	43.5		
		MCWB	37.5	36.0	41.6	47.6	54.1	58.9	63.8	61.7	57.8	49.8	43.4	36.5		
	5%	DB	41.6	39.5	49.6	60.9	73.6	75.4	79.9	80.7	77.3	61.3	49.2	40.0		
		MCWB	35.6	33.8	39.1	44.0	52.2	57.2	63.2	61.0	56.2	47.3	40.0	33.7		
	10%	DB	38.7	35.8	44.4	55.7	69.1	72.1	76.7	76.7	71.9	56.4	43.9	36.5		
		MCWB	33.4	31.4	36.8	41.7	50.3	55.7	61.8	60.1	54.3	44.9	36.4	31.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.5	40.7	46.9	51.3	56.6	64.2	70.2	66.6	60.7	54.8	47.7	40.1		
		MCDB	47.6	47.9	60.1	70.3	78.3	78.0	81.0	78.8	81.2	72.1	61.4	46.6		
	2%	WB	37.7	36.5	42.9	48.0	54.8	61.5	66.5	64.6	58.9	51.4	43.7	36.6		
		MCDB	43.7	42.5	52.9	65.3	74.3	74.4	79.0	78.1	77.9	63.3	54.0	43.2		
	5%	WB	35.8	34.1	39.8	45.1	53.2	59.3	64.6	62.8	56.8	48.3	40.6	33.9		
		MCDB	41.3	39.2	48.6	58.7	71.0	70.9	76.8	76.4	74.6	60.0	48.5	39.8		
	10%	WB	33.5	31.6	36.9	42.5	51.3	57.3	62.8	61.1	54.9	45.6	37.2	31.4		
		MCDB	38.5	35.4	43.8	54.1	65.9	69.1	74.2	73.9	70.4	55.8	43.1	36.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	16.0	17.5	20.9	22.8	21.1	22.8	23.5	23.9	20.5	16.7	15.0		
		MCDBR	16.5	19.0	24.8	30.1	29.7	26.4	28.1	30.6	33.8	27.6	24.1	17.4		
		MCWBR	11.8	13.4	14.1	15.0	12.3	11.5	13.6	12.7	14.0	14.2	14.5	12.4		
	5% WB	MCDBR	16.6	18.5	23.5	28.2	26.5	23.4	25.4	26.5	31.2	26.1	23.5	18.2		
		MCWBR	12.0	13.3	14.2	14.7	11.7	11.6	13.4	12.5	13.5	13.7	14.7	13.5		
(40) Clear-Sky Solar Irradiance	taub		0.259	0.267	0.281	0.337	0.359	0.351	0.350	0.368	0.322	0.281	0.252	0.238		
	taud		2.378	2.437	2.475	2.327	2.317	2.386	2.406	2.346	2.441	2.535	2.491	2.399		
	Ebn at Noon		248	280	296	286	284	286	285	273	276	272	253	241		
	Edh at Noon		19	23	28	36	38	36	35	35	29	21	17	16		
(44) All-Sky Solar Radiation	RadAvg		358	680	1112	1512	1823	1964	2109	1701	1199	728	391	278		
	RadStd		27	58	92	115	123	119	104	118	121	84	33	27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (78 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page